

CSC 1031 - Movie Ticket Booking System

4.1 Cover Page

Project Title : Movie Ticket Booking System
Course Name : CSC 1031
Team Name : Group X-MovieBookingSystem
Submission Date : 24th July 2026

Team Members:

- WGJC SANDARUWAN - (AS20250395)
- GDC RASHMIKA - (AS20250586)
- WPN PIYUMAL - (AS20250558)
- DN SAMARAJEEWA - (AS20250623)
- TVDR SANDARUWAN - (AS20250627)

4.2 Introduction

Managing theater bookings manually can lead to double-booked seats and incorrect price calculations. The purpose of this project was to build a command-line Movie Ticket Booking System using the C programming language to automate these processes.

Our system allows theater staff to view showtimes, check seat availability via a visual seat map, book seats for customers, apply appropriate discounts (Student, Senior, or Group), cancel bookings, and view a real-time revenue report. All data is managed in-memory during the program's execution.

4.3 Team Contribution Summary

Team Member	Assigned Module	Specific Contribution
WGJC SANDARUWAN	main.c & booking.h	Set up the GitHub repository, created the main menu loop, integrated all modules, and defined the core structs (Movie, Seat, Booking) in the header file.

Team Member	Assigned Module	Specific Contribution
TVDR SANDARUWAN	display.c	Developed the UI elements and the logic to print the 2D seat map grid, correctly formatting free (.) and booked (X) seats.
WPN PIYUMAL	reserve.c	Handled the core booking logic. Added input validation to prevent booking out-of-range or already taken seats, and implemented the cancellation function.
GDC RASHMIKA	pricing.c	Implemented the calculation rules based on seat tiers (Regular, Premium, VIP) and combined the logic for Student, Senior, and Group discounts without overlapping restricted categories.
DN SAMARAJEEWA	Documentation	Compiled the project report, tracked GitHub issues, gathered screenshots of the running program, and formatted the final submission files.

4.4 Design Overview

Data Structures: We used several structs in booking.h to organize our data. The Seat struct holds the row, column, tier (Regular, Premium, VIP), and a boolean flag indicating if it is booked. The Booking struct stores the customer's name, seat reference, and the final price paid. The Movie struct holds the movie name and a 2D array of Seats representing the showtimes.

Module Connections:

- main.c acts as the controller. It includes booking.h and calls functions from other files based on user input.
- display.c reads the arrays to print outputs.
- reserve.c updates the boolean flags in the Seat structs.
- pricing.c is called by reserve.c during the booking process to return the final integer value for the ticket price.

4.5 Screenshots & Program Walkthrough

Screenshot 1: The Main Menu

```
-----***** Movie Ticket Booking System *****-----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit
Enter your choice: |
```

```
-----***** Movie Ticket Booking System *****-----
1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit
Enter your choice: 1

=====
AVAILABLE MOVIES & SHOWTIMES
=====

Movie 1: Movie 1
-----
[1] Showtime: 14:00 PM
[2] Showtime: 15:00 PM

Movie 2: Movie 2
-----
[1] Showtime: 14:00 PM
[2] Showtime: 15:00 PM

Movie 3: Movie 3
-----
[1] Showtime: 14:00 PM
[2] Showtime: 15:00 PM
=====
```

Screenshot 2: Seat Map before any booking

```
Enter your choice: 2
Enter Movie Number (1-3): 1
Enter Showtime Number (1-2): 1

=====
SEAT MAP
=====

   1  2  3  4  5  6  7  8  9  10
-----
Row A | . | . | . | . | . | . | . | . | . | <- Regular (Rs. 500)
-----
Row B | . | . | . | . | . | . | . | . | . | <- Regular (Rs. 500)
-----
Row C | . | . | . | . | . | . | . | . | . | <- Premium (Rs. 750)
-----
Row D | . | . | . | . | . | . | . | . | . | <- Premium (Rs. 750)
-----
Row E | . | . | . | . | . | . | . | . | . | <- VIP      (Rs. 1000)
-----

Legend: [ . ] = Free Seat | [ X ] = Booked Seat
=====
```

Screenshot 3: Successful seat booking

```
-----***** Movie Ticket Booking System *****-----

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit

Enter your choice: 3
Enter Movie Number (1-3): 2
Enter Showtime Number (1-2): 2

=====
                BOOK A SEAT
=====
Enter Row (A - E): A
Enter Seat Number (1 - 10): 4
Enter Customer Name: Danith

Select Discount Category:
0 = None
1 = Student (10% off)
2 = Senior Citizen (20% off)
Enter category (0, 1, or 2): 1
How many tickets are you booking in total? (For Group Discount): 1

Success! Seat A4 successfully booked for Danith.
Ticket Price: LKR 450.00
```

```
-----***** Movie Ticket Booking System *****-----

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit

Enter your choice: 3
Enter Movie Number (1-3): 1
Enter Showtime Number (1-2): 1

=====
                BOOK A SEAT
=====
Enter Row (A - E): E
Enter Seat Number (1 - 10): 3
Enter Customer Name: Deshan

Select Discount Category:
0 = None
1 = Student (10% off)
2 = Senior Citizen (20% off)
Enter category (0, 1, or 2): 1
How many tickets are you booking in total? (For Group Discount): 2

Success! Seat E3 successfully booked for Deshan.
Ticket Price: LKR 900.00
```

```

-----***** Movie Ticket Booking System *****-----

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit

Enter your choice: 3
Enter Movie Number (1-3): 1
Enter Showtime Number (1-2): 1

=====
                     BOOK A SEAT
=====
Enter Row (A - E): A
Enter Seat Number (1 - 10): 1
Enter Customer Name: Nimeth Piyumal

Select Discount Category:
0 = None
1 = Student (10% off)
2 = Senior Citizen (20% off)
Enter category (0, 1, or 2): 0
How many tickets are you booking in total? (For Group Discount): 1

Success! Seat A1 successfully booked for Nimeth Piyumal.
Ticket Price: LKR 500.00

```

Screenshot 4: Attempting to book an already-taken seat

```

-----***** Movie Ticket Booking System *****-----

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit

Enter your choice: 3
Enter Movie Number (1-3): 1
Enter Showtime Number (1-2): 1

=====
                     BOOK A SEAT
=====
Enter Row (A - E): E
Enter Seat Number (1 - 10): 3

Seat E3 is ALREADY BOOKED by Deshan!

```

Screenshot 5: Booking cancellation

```

-----***** Movie Ticket Booking System *****-----

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit

Enter your choice: 4
Enter Movie Number (1-3): 1
Enter Showtime Number (1-2): 1

=====
                     CANCEL BOOKING
=====
Enter Row (A - E): A
Enter Seat Number (1 - 10): 1

Success! Booking for Seat A1 has been CANCELLED.
Refunded Amount: LKR 500.00

```

Screenshot 6: Search result for a booking

```
-----***** Movie Ticket Booking System *****-----

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit

Enter your choice: 5

--- Search Booking ---
1. Search by Customer Name
2. Search by Seat Number
Enter your choice: 1
Enter Customer Name: Deshan

Booking Found!
Movie: Movie 1
Showtime: 14:00 PM
Seat: E3
Price Paid: Rs. 900.00
```

```
-----***** Movie Ticket Booking System *****-----

1. View Showtimes
2. View Seat Map
3. Book a Seat
4. Cancel a Booking
5. Search Booking
6. View Revenue Report
7. Exit

Enter your choice: 5

--- Search Booking ---
1. Search by Customer Name
2. Search by Seat Number
Enter your choice: 2
Enter Seat Row (A-E): A
Enter Seat Number (1-10): 4

Booking Found in Movie 2 (15:00 PM)!
Customer Name: Danith
Price Paid: Rs. 450.00
```

Screenshot 7: Revenue report screen

```
--- Revenue Report ---

Movie: Movie 1
  Showtime: 14:00 PM
  Tickets Sold: 1
  Revenue: Rs. 900.00
  Showtime: 15:00 PM
  Tickets Sold: 0
  Revenue: Rs. 0.00

Movie: Movie 2
  Showtime: 14:00 PM
  Tickets Sold: 0
  Revenue: Rs. 0.00
  Showtime: 15:00 PM
  Tickets Sold: 1
  Revenue: Rs. 450.00

Movie: Movie 3
  Showtime: 14:00 PM
  Tickets Sold: 0
  Revenue: Rs. 0.00
  Showtime: 15:00 PM
  Tickets Sold: 0
  Revenue: Rs. 0.00

=====
Total Tickets Sold (All Movies): 2
Total Revenue (All Movies): Rs. 1350.00
=====
```

4.6 Challenges & How They Were Solved

- **Challenge 1: Passing 2D Arrays between Functions**

Initially, we faced segmentation faults (crashes) when trying to pass our 2D seat map array from main.c to the display functions in display.c. We got confused with C pointers and array decay.

- *Solution:* We resolved this by explicitly defining the column size in our function prototypes in booking.h (e.g., void displayMap(Seat seats[][10], int rows);) which allowed the compiler to correctly calculate memory offsets without crashing.

- **Challenge 2: Overlapping Discount Rules**

The assignment stated that Student and Senior discounts cannot be combined, but the Group discount CAN be combined with them. Early in testing, our pricing.c module was applying 30% off for a student group (10% + 20%), which was calculating incorrectly against the base price sequentially.

- *Solution:* We restructured the logic to first establish the "Base Discount" (either Student 10%, Senior 20%, or Regular 0%). Then, we checked the group size. If it was ≥ 4 , we applied the 10% Group discount to the *newly discounted* amount, ensuring accurate real-world revenue calculations.

- **Challenge 3: GitHub Merge Conflicts**

Since this was our first time collaborating heavily on GitHub, we ran into merge conflicts when two members accidentally modified main.c at the same time on different branches.

- *Solution:* We held a quick Discord call, carefully reviewed the conflicting lines in GitHub's web editor, kept both necessary changes, and committed the resolved merge. We then agreed to clearly communicate in our WhatsApp group before modifying core files.

4.7 GitHub Collaboration Summary

Our team used a feature-branch workflow. WGJC SANDARUWAN set up the main repository, and the rest of the team cloned it and created branches for their specific '.c' files. We used GitHub Issues to track who was doing what module.

Issues

Assigned to me

Created by me

Mentioned

Recent activity

Views

Projects

Milestones

Labels

Feedback

Preview

Collapse sidebar

All issues

New issue

issue state:open

Open5

Closed0

AuthorLabelsProjectsMilestonesAssigneesF1 Newest

Interface of project

#7 - janithchamika opened 3m ago

Documentation setup

#6 - janithchamika opened 4m ago

Develop pricing logic

#5 - janithchamika opened 5m ago

Create Reserver module

#4 - janithchamika opened 6m ago

Create display module

#3 - janithchamika opened 7m ago

[Give feedback](#)

Commits

main

All usersAll time

Commits on Jul 24, 2026

How to RUN this code

janithchamika authored 1 hour ago

Verified09f0669

Added initial main.c skeleton code - DISPLAY MENU

janithchamika committed 9 hours ago

f5b4307

How to Run this Code

janithchamika committed 9 hours ago

29dff77

Add booking.h header file

janithchamika committed 10 hours ago

d5db622

Commits on Jul 20, 2026

Added initial main.c skeleton code - DISPLAY MENU

janithchamika committed 4 days ago

bb8a06e

Commits on Jul 18, 2026

Initial commit

janithchamika authored last week

Verified52d0b54

Commits

feature-display

All usersAll time

Commits on Jul 24, 2026

Add display module for seat map

deshanrashmika589 committed 2 hours ago

ef51c2c

Added initial main.c skeleton code - DISPLAY MENU

janithchamika committed 9 hours ago

f5b4307

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bb8a06e

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Initial commit

janithchamika authored last week

Verified52d0b54

Commits

feature-pricing

All usersAll time

Commits on Jul 24, 2026

Add pricing module for booking.h

chamathrashmika39-collab committed 2 hours ago

9e387eb

Added initial main.c skeleton code - DISPLAY MENU

janithchamika committed 9 hours ago

f5b4307

How to Run this Code

janithchamika committed 9 hours ago

29dff77

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janithchamika committed 4 days ago

bb8a06e

Commits on Jul 18, 2026

Initial commit

janithchamika authored last week

Verified

52d0b54

Commits

feature-reserve-

All usersAll time

Commits on Jul 24, 2026

Create reserve.c

nimethpiyumal committed 3 hours ago

cdceda2

Added initial main.c skeleton code - DISPLAY MENU

janithchamika committed 9 hours ago

f5b4307

How to Run this Code

janithchamika committed 9 hours ago

29dff77

Add booking.h header file

janithchamika committed 10 hours ago

d5db622

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janithchamika committed 4 days ago

bb8a06e

Commits on Jul 18, 2026

Initial commit

janithchamika authored last week

Verified

52d0b54

Commits

feature-documen...

All usersAll time

Commits on Jul 24, 2026

add final submission files

danithnethsara-oss committed 3 minutes ago

f253df9

How to RUN this code

janithchamika authored 1 hour ago

Verified

09f0669

Added initial main.c skeleton code - DISPLAY MENU

janithchamika committed 9 hours ago

f5b4307

How to Run this Code

janithchamika committed 9 hours ago

29dff77

Add booking.h header file

janithchamika committed 10 hours ago

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52d0b54